▶ Virtual Memory

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Virtual Memory

- **Virtual Memory** is a technique that allows the execution of processes that are not completely in main memory.
- **Motivation**
  - Conditional execution:
    - either 2 pages of **if** block will be loaded

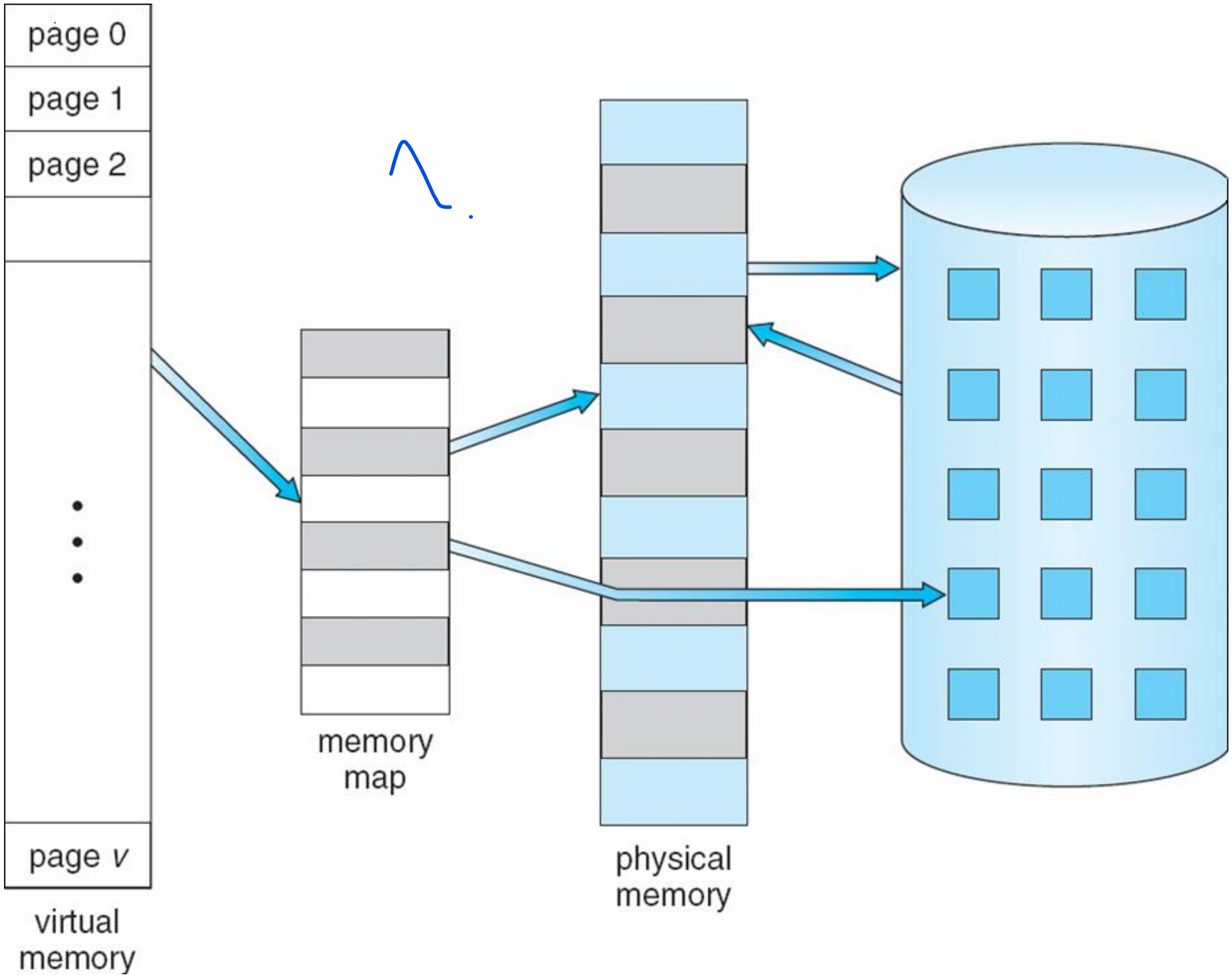
      or 2 pages of **else** block.
  - Error handling codes in programs are seldom executed.
  - Even though all the pages are needed, they are not needed at the same time.

```
If(condition){
  ----
 |    |
  ----
 |    |
  ----
}
Else{
  ----
 |    |
  ----
 |    |
  ----
}
```

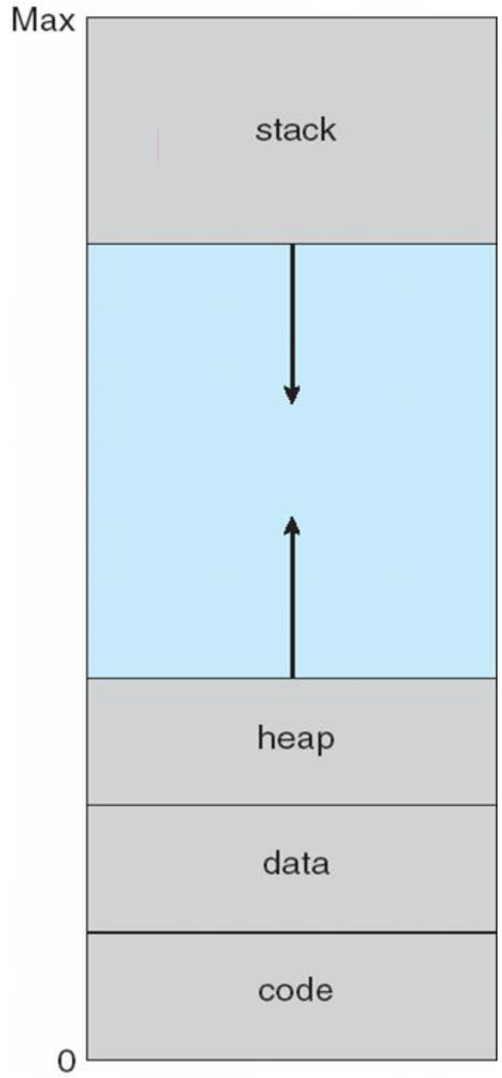
Virtual Memory

- Advantages:
  - A process would no longer be constrained by the amount of available physical memory.
  - Increases degree of multi-programming.
  - Because each process is only using a fraction of their total address space, there is more memory left for other programs, improving CPU utilization and system throughput.
  - Less I/O is needed for swapping processes in and out of RAM, speeding things up.

Virtual Memory That is Larger Than Physical Memory



Virtual-address Space



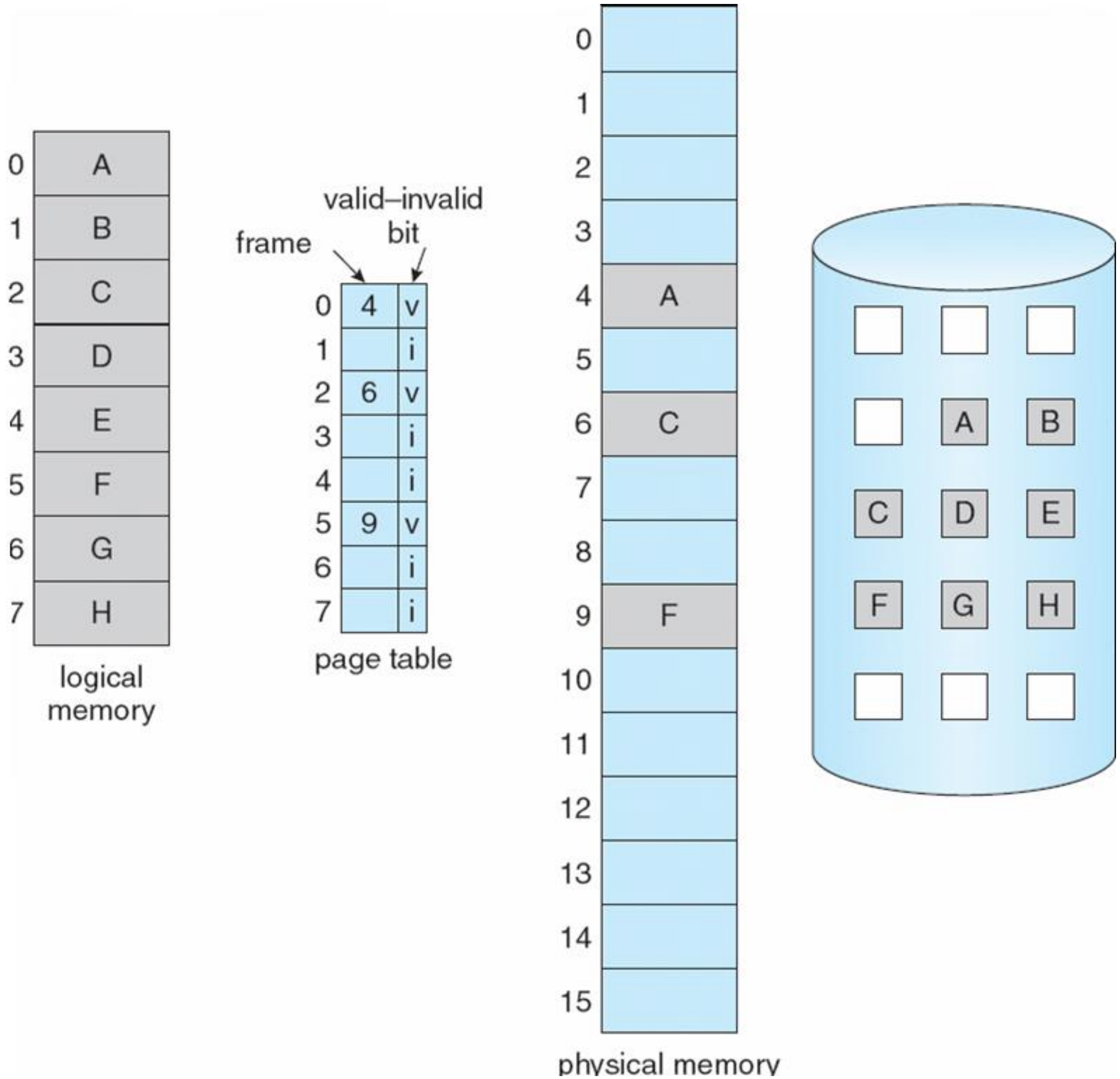
Demand Paging

- The basic idea behind ***demand paging*** is that when a process is swapped in, its pages are not swapped in all at once.
- Rather they are swapped in only when the process needs them. (on demand.) This is termed a ***lazy swapper***, although a ***pager*** is a more accurate term.
- Bring a page into memory only when it is needed.
- Page is needed $\Rightarrow$ reference to it
  - invalid reference (not-in-memory) $\Rightarrow$ bring to memory

Valid-Invalid Bit

- With each page table entry a valid–invalid bit is associated (**1** $\Rightarrow$ in-memory, **0** $\Rightarrow$ not-in-memory)
- Initially valid–invalid bit is set to **0** on all entries
- During address translation, if valid–invalid bit in page table entry is **0** $\Rightarrow$ page fault

Page Table When Some Pages Are Not in Main Memory



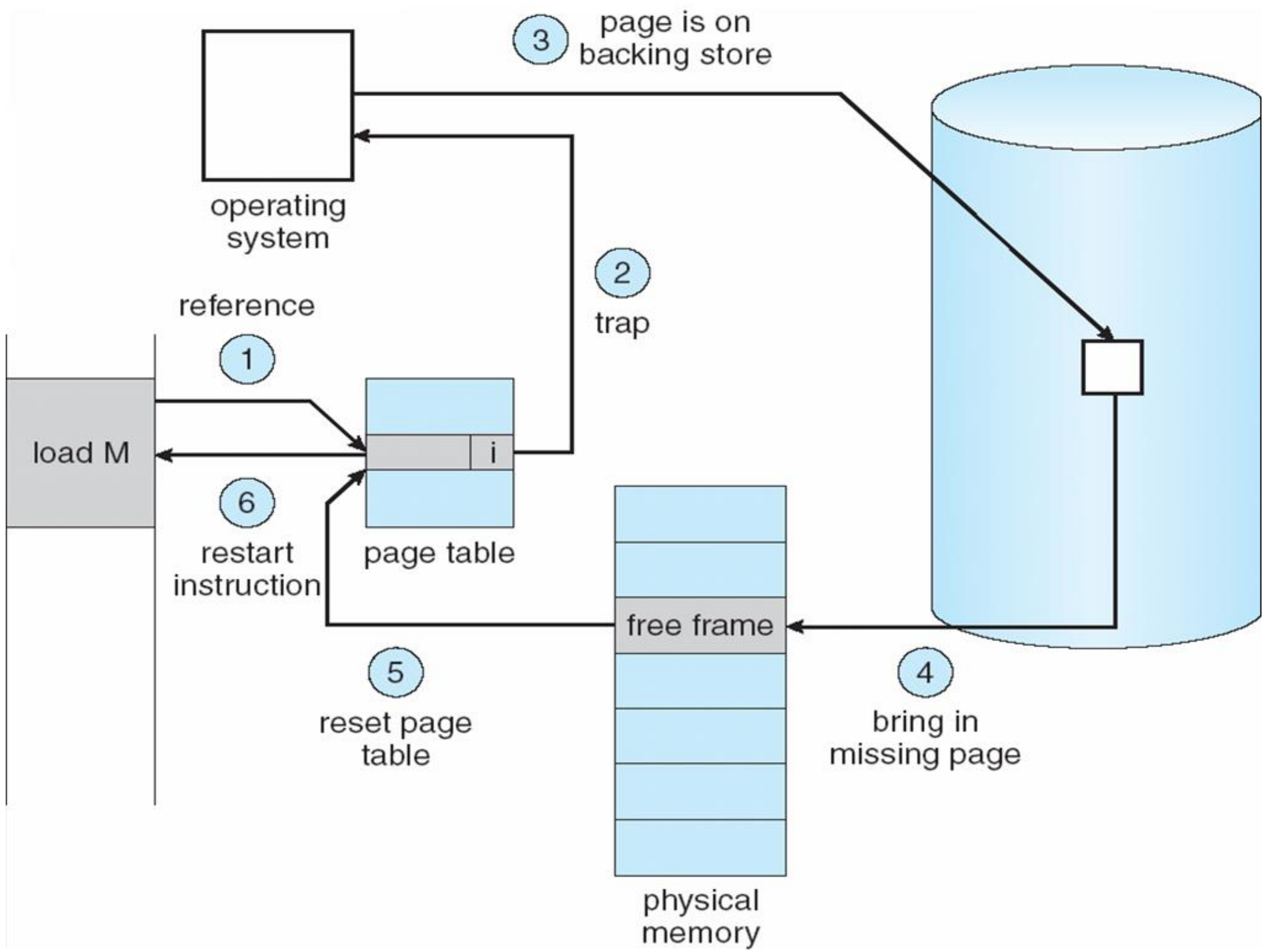
Valid-Invalid Bit

if inv , process terminated else free frame located and disc op bring in page from disc blocking the process on io wait then page table updated and instructiion that caused fault rr exec

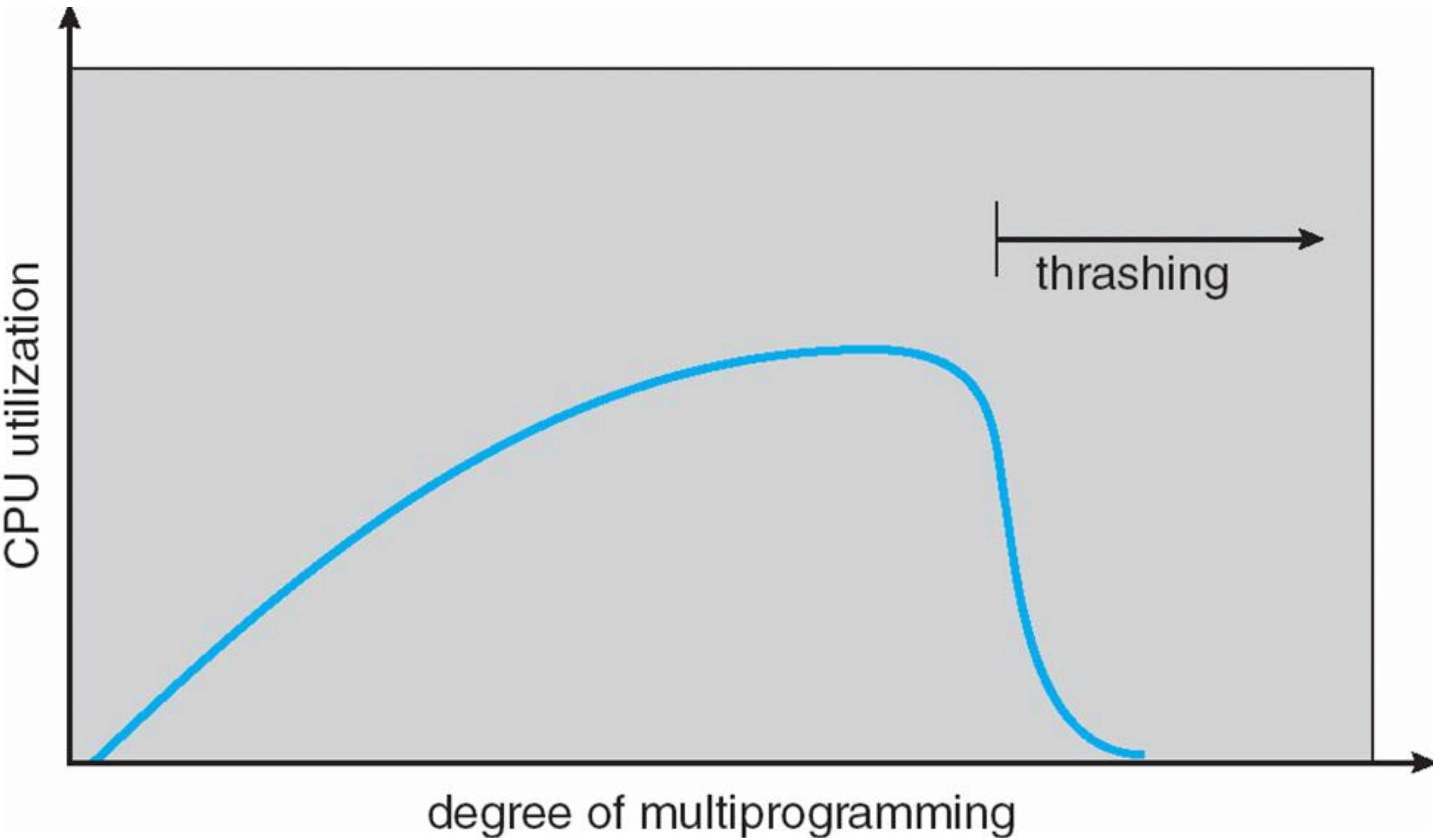
On the other hand, if a page is needed that was not originally loaded up, then a ***page fault trap*** is generated, which must be handled in a series of steps:

- The memory address requested is first checked, to make sure it was a valid memory request.
- If the reference was invalid, the process is terminated. Otherwise, the page must be paged in.
- A free frame is located, possibly from a free-frame list.
- A disk operation is scheduled to bring in the necessary page from disk. (This will usually block the process on an I/O wait, allowing some other process to use the CPU in the meantime.)
- When the I/O operation is complete, the process's page table is updated with the new frame number, and the invalid bit is changed to indicate that this is now a valid page reference.
- The instruction that caused the page fault must now be restarted from the beginning, (as soon as this process gets another turn on the CPU.)

Steps in Handling a Page Fault



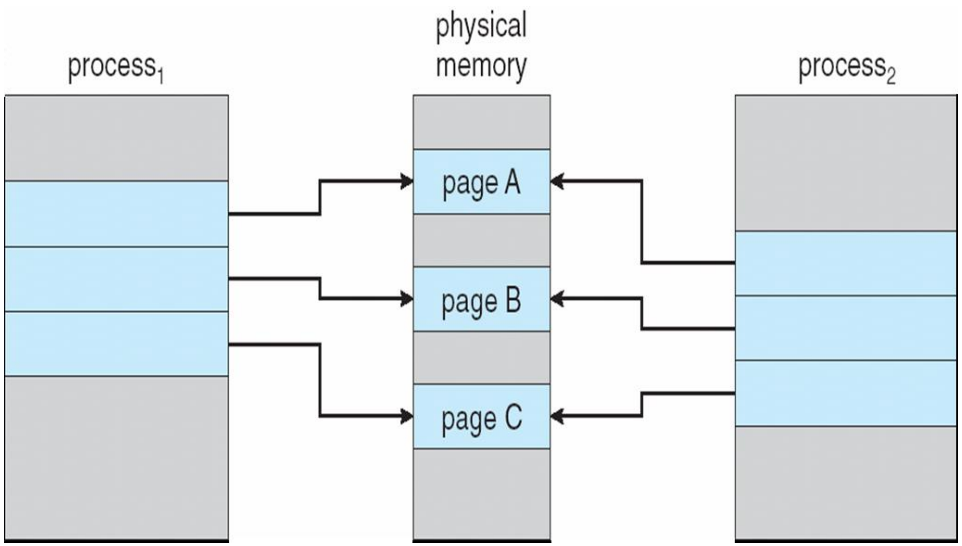
Thrashing



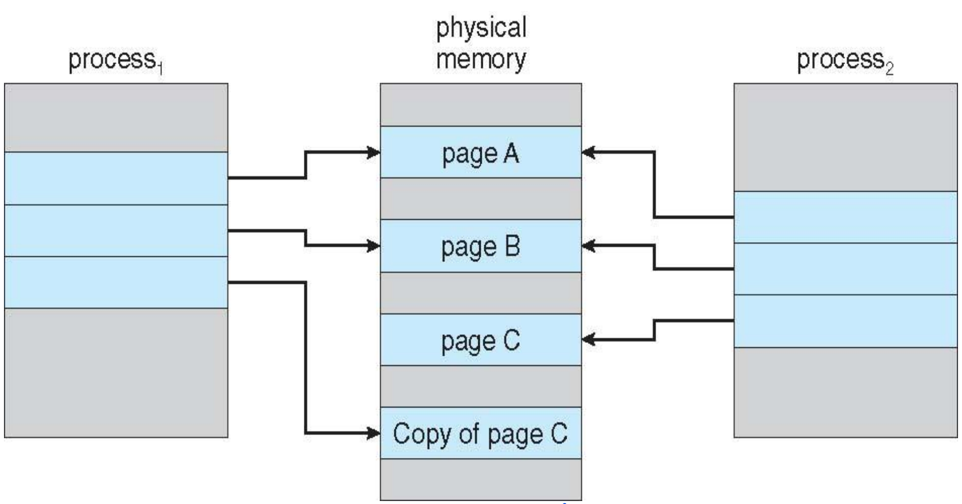
Copy-on-Write

- Copy-on-Write allows both parent and child processes to initially *share* the same pages in memory
- If either process modifies a shared page, only then is the page copied
- It allows more efficient process creation as only modified pages are copied

Copy-on-Write



Before Process 1 Modifies Page C

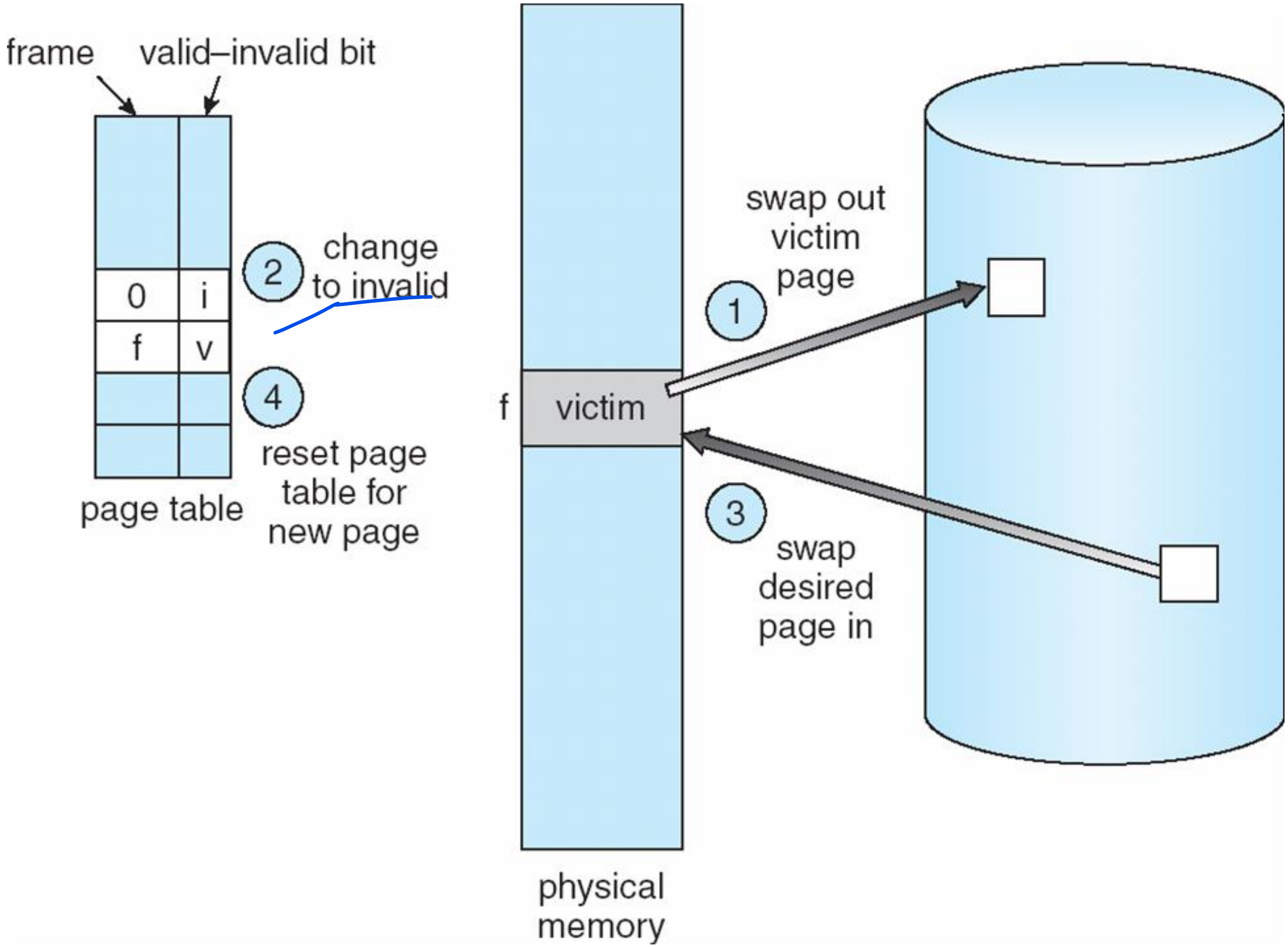


After Process 1 Modifies Page C

What happens if there is no free frame?

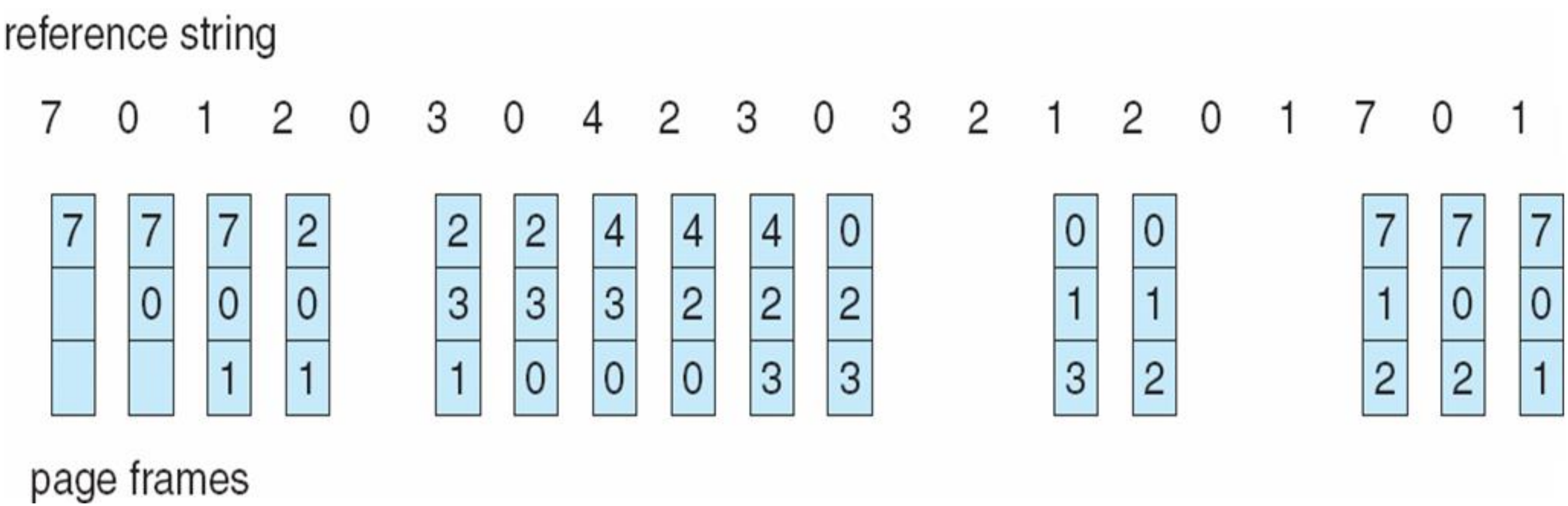
- Page replacement – find some page in memory, but not really in use, swap it out
  - algorithm
  - performance – want an algorithm which will result in minimum number of page faults
  - Use **modify (dirty) bit** to reduce overhead of page transfers – only modified pages are written to disk

Page Replacement



First-In-First-Out (FIFO) Algorithm

Replace page that has come first.



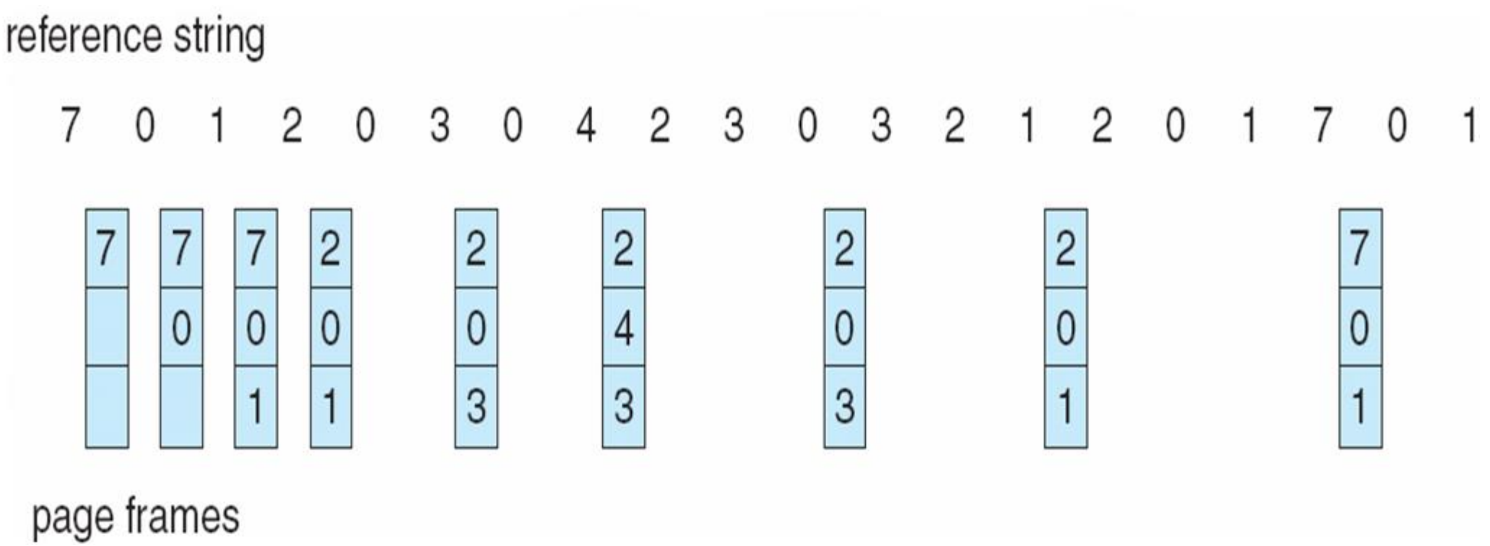
Total page faults = 15

FIFO Algorithm: Belady's Anomaly

- Reference string: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5
- 3 frames – 9 page faults
- 4 frames – 10 page faults

Optimal Algorithm

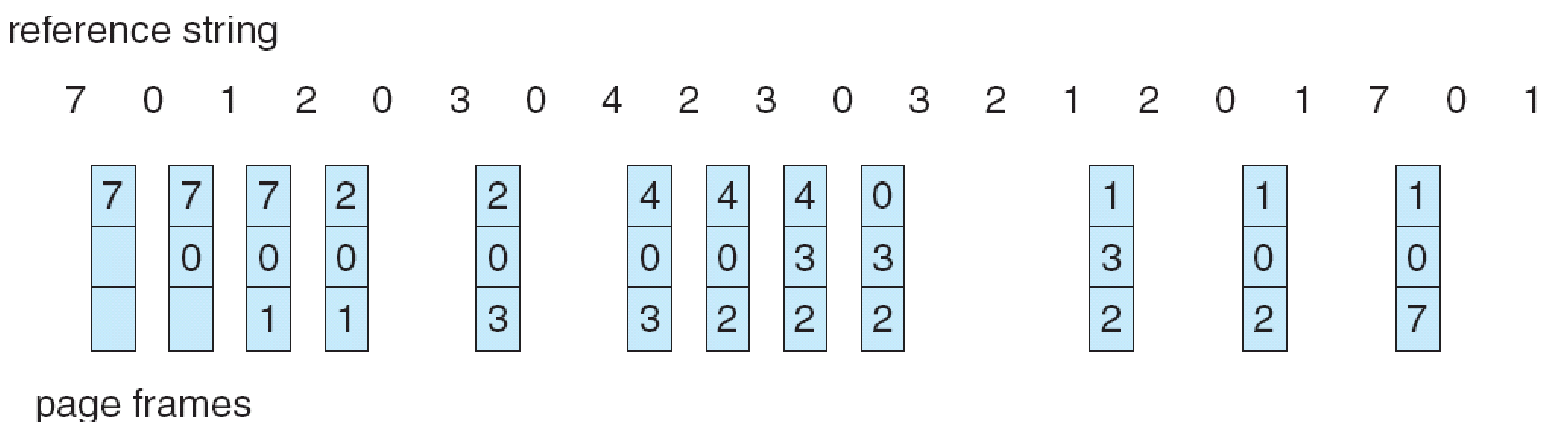
- Replace page that will not be used for longest period of time



- How do you know this?
- Used for measuring how well your algorithm performs

Least Recently Used (LRU) Algorithm

- Replace the page that has not been used for longest period of time.



LRU Implementation

- Counter implementation
  - Every page entry has a counter; every time page is referenced, copy the clock into the counter
  - Logical counter ticks with every page reference
  - Replace a valid page with smallest counter

LRU Implementation

- Modified Stack implementation
  - Push at top only, Pop from anywhere
  - Page referenced $\Rightarrow$ on the memory $\Rightarrow$ move it to the top
  - LRU page will be at the bottom always

reference string

4 7 0 7 1 0 1 2 1 2 7 1 2

Additional Reference bits Algorithm

- Keep a reference/use bit and 7-bit reference history for each entry in PMT.
- For every page reference, update the reference bit
- At regular intervals (say 50ns)
  - Perform right shift to the history bits, discarding the LSB
  - Push reference bit value to MSB of history bits
  - Reset reference bit to 0
- Thus 1-byte represents the status of the page usage in 8 intervals.
  - 00000000 – The page has not been used for 8 intervals
  - 11111111 – The page is used at least once in each interval
  - 11000100 is more recently used than 01110111
  - So, lowest number is the LRU page

Additional Reference bits Algorithm

3, 2, 3, T, 8, 0, 3, T, 3, 0, 2, T, 6, 3, 4, 7

each interval one shift

T marks the end of each time interval

Initial state

During 1st interval

P	U3	U2	U1	U0
3	1	0	0	0
2	1	0	0	0
-	0	0	0	0
-	0	0	0	0
-	0	0	0	0

At the end of 1st interval

P	U3	U2	U1	U0
3	0	1	0	0
2	0	1	0	0
-	0	0	0	0
-	0	0	0	0
-	0	0	0	0

Additional Reference bits Algorithm

During 2nd interval

P	U3	U2	U1	U0
3	1	1	0	0
2	0	1	0	0
8	1	0	0	0
0	1	0	0	0
-	0	0	0	0

At the end of 2nd interval

P	U3	U2	U1	U0
3	0	1	1	0
2	0	0	1	0
8	0	1	0	0
0	0	1	0	0
-	0	0	0	0

Additional Reference bits Algorithm

During 3rd interval

P	U3	U2	U1	U0
3	1	1	1	0
2	1	0	1	0
8	0	1	0	0
0	1	1	0	0
-	0	0	0	0

At the end of 3rd interval

P	U3	U2	U1	U0
3	0	1	1	1
2	0	1	0	1
8	0	0	1	0
0	0	1	1	0
-	0	0	0	0

Additional Reference bits Algorithm

During 4th interval, after 6, 3

P	U3	U2	U1	U0
3	1	1	1	1
2	0	1	0	1
8	0	0	1	0
0	0	1	1	0
6	1	0	0	0

During 4th interval, after 6, 3, 4

P	U3	U2	U1	U0
3	1	1	1	1
2	0	1	0	1
4	1	0	0	0
0	0	1	1	0
6	1	0	0	0

Page 4 replaced 8 with U bits 0010

Additional Reference bits Algorithm

During 4th interval, after 6, 3, 4, 7

P	U3	U2	U1	U0
3	1	1	1	1
7	1	0	0	0
4	1	0	0	0
0	0	1	1	0
6	1	0	0	0

If there are more victim pages with same U bits then chose FIFO

Page 7 replaced 2 with U bits 0101

Second Chance Algorithm

- Uses
  - a reference/used bit
  - NO history bits
  - FIFO

```
Search FIFO pages circularly
   If FIFO page's used bit is 0 then
        replace it
   Else
        give that FIFO page a second chance
        clear its used bit
        goto next FIFO page
```

Second Chance Algorithm

```
   P    U    3      P    U    2     P    U    3
 +----+----+      +----+----+     +----+----+
 |    | 0  |*     | 3  | 1  |     | 3  | 1  |
 +----+----+      +----+----+     +----+----+
 |    | 0  |      |    | 0  |*    | 2  | 1  |
 +----+----+      +----+----+     +----+----+
 |    | 0  |      |    | 0  |     |    | 0  |*
 +----+----+      +----+----+     +----+----+
 |    | 0  |      |    | 0  |     |    | 0  |
 +----+----+      +----+----+     +----+----+
 |    | 0  |      |    | 0  |     |    | 0  |
 +----+----+      +----+----+     +----+----+
```

Second Chance Algorithm

```
   P     U     0     P     U     8     P     U    4
+----+----+       +----+----+       +----+----+
|  3 |  1 |       |  3 |  1 |       |  3 |  1 |
+----+----+       +----+----+       +----+----+
|  2 |  1 |       |  2 |  1 |       |  2 |  1 |
+----+----+       +----+----+       +----+----+
|    |  0 | *     |  0 |  1 |       |  0 |  1 |
+----+----+       +----+----+       +----+----+
|    |  0 |       |    |  0 | *     |  8 |  1 |
+----+----+       +----+----+       +----+----+
|    |  0 |       |    |  0 |       |    |  0 | *
+----+----+       +----+----+       +----+----
```

Second Chance Algorithm

```
  P   U   2    P   U   5   P   U   0
+---+---+    +---+---+   +---+---+
| 3 | 1 |*   | 3 | 1 |*  | 5 | 1 |
+---+---+    +---+---+   +---+---+
| 2 | 1 |    | 2 | 1 |   | 2 | 0 |*
+---+---+    +---+---+   +---+---+
| 0 | 1 |    | 0 | 1 |   | 0 | 0 |
+---+---+    +---+---+   +---+---+
| 8 | 1 |    | 8 | 1 |   | 8 | 0 |
+---+---+    +---+---+   +---+---+
| 4 | 1 |    | 4 | 1 |   | 4 | 0 |
+---+---+    +---+---+   +---+---+
```

Second Chance Algorithm

```
  P    U     9    P    U     8    P    U    3
+----+----+     +----+----+     +----+----+
| 5  | 1  |     | 5  | 1  |     | 5  | 1  |
+----+----+     +----+----+     +----+----+
| 2  | 0  |*    | 9  | 1  |     | 9  | 1  |
+----+----+     +----+----+     +----+----+
| 0  | 1  |     | 0  | 1  |*    | 0  | 1  |*
+----+----+     +----+----+     +----+----+
| 8  | 0  |     | 8  | 0  |     | 8  | 1  |
+----+----+     +----+----+     +----+----+
| 4  | 0  |     | 4  | 0  |     | 4  | 0  |
+----+----+     +----+----+     +----+----
```

Second Chance Algorithm

```
   P    U    2      P    U
+----+----+      +----+----+
| 5  | 1  |*     | 5  | 0  |
+----+----+      +----+----+
| 9  | 1  |      | 9  | 0  |
+----+----+      +----+----+
| 0  | 0  |      | 2  | 1  |
+----+----+      +----+----+
| 8  | 0  |      | 8  | 0  |*
+----+----+      +----+----+
| 3  | 1  |      | 3  | 1  |
+----+----+      +----+----+
```

Enhanced Second Chance Algorithm

- Considers reference/used bit and dirty/modified bit
- Four cases, based on (R,M), the pair:
  - (R,M) = (0,0) neither recently used nor modified -- best to replace
  - (R,M) = (0,1) not recently used but modified -- second choice
  - (R,M) = (1,0) recently used but not modified -- third choice
  - (R,M) = (1,1) recently used and modified -- fourth choice

Enhanced Second Chance Algorithm

- If class-1 page then use it
- If class-2 page then record first instance
- Else clear R bit and go to next FIFO page
- If scan is completed then
  - If class-2 page was recorded then use that recorded instance
  - Else repeat another scan

Counting based page replacements

Keep a counter of the number of references that have been made to each page

- **LFU Algorithm**: replaces page with smallest count
- **MFU Algorithm**: based on the argument that the page with the smallest count was probably just brought in and has yet to be used

Allocation of Frames

- Each process needs *minimum* number of frames
  - Otherwise high page fault rate => slow execution
- Two major allocation schemes
  - Fixed allocation
  - Priority allocation

Fixed Allocation: Equal allocation

- Equal allocation – For example, if there are m=100 frames and n=5 processes, give each process 20 frames.
  - Smaller processes waste frames.
- Proportional allocation – Allocate according to the size of process

$$s_i = \text{size of process } p_i$$

$$S = \sum s_i$$

$$m = \text{total number of frames}$$

$$a_i = \text{allocation for } p_i = \frac{s_i}{S} \times m$$

$$m = 64$$

$$s_i = 10$$

$$s_2 = 127$$

$$a_1 = \frac{10}{137} \times 64 \approx 5$$

$$a_2 = \frac{127}{137} \times 64 \approx 59$$

Priority Allocation

- Use a proportional allocation scheme using priorities rather than size

Global vs. Local Allocation

- **Global replacement** – process selects a replacement frame from the set of all frames; one process can take a frame from another
- **Local replacement** – each process selects from only its own set of allocated frames

- In local replacement, no. of allocated frames is not affected by other processes => page fault rate is not affected by other processes.

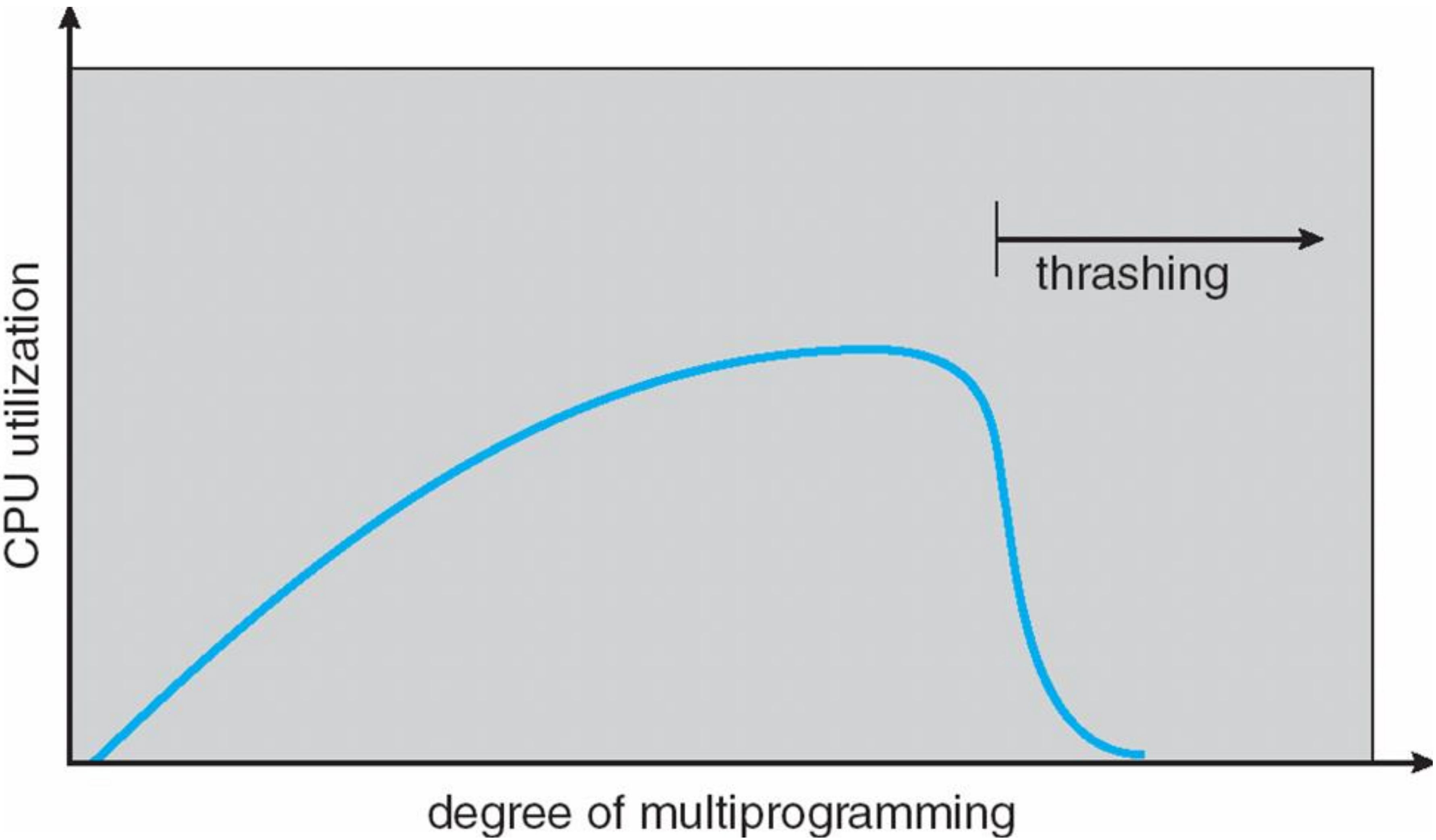
Thrashing

- P_{high} steals frames from P_{low}
- Page fault rate increases for P_{low}
- P_{high} steals more frames from P_{low}
- Page fault rate further increases for P_{low}

- If a process does not have "enough" frames, the page-fault rate is very high. This leads to:
  - low CPU utilization
  - operating system thinks that it needs to increase the degree of multiprogramming
  - another process added to the system

- **Thrashing** $\equiv$ a process is busy swapping pages in and out

Thrashing



Demand Paging and Thrashing

Why does demand paging work?

Locality model

- Process migrates from one locality to another
  - Currently executing **main** function (locality-1), may call **sqrt** function (locality-2)
- Localities may overlap
  - both the functions may use the same global pages

Why does thrashing occur?

- $\sum$no. of pages actively used > total memory size

Solution to Thrashing

- Use local page replacement algorithm.
- Use working-set model.
- Page fault frequency scheme.

Working-set Model

- **Working set**: collection of pages that a process is **working** with, i.e., the set of pages that a process is actively using.
- The working set must be memory-resident to prevent this process from thrashing.
- $\Delta \equiv$ working-set window $\equiv$ a fixed number of page references (say 10)
- $WSS_i \equiv$ working set size of Process P_i
- $D = \Sigma\ WSS_i \equiv$ total demand for frames
- if $D > m \Rightarrow$ Thrashing
- Policy if D > m, then suspend one of the processes

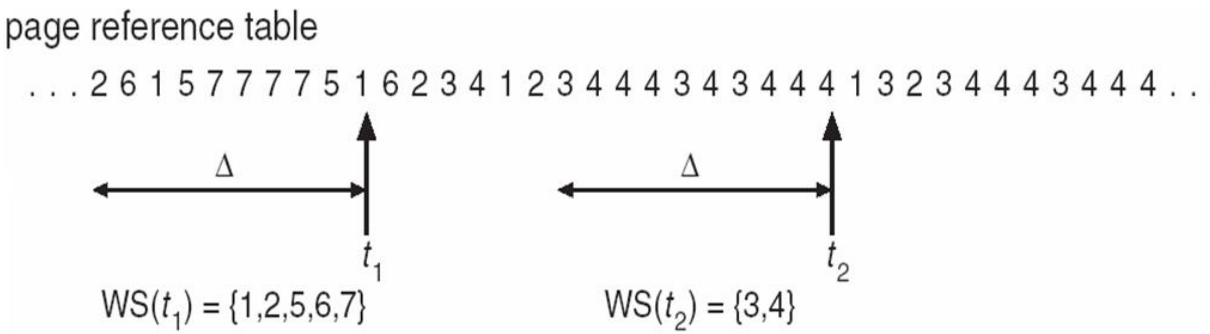
Working-set Model

- How to decide working set?
  - Working set window (Δ)
  - Idle page time (T)

Working-set Model

Working set window (Δ)

- $\Delta \equiv$ a fixed number of previous page references (say 10)



- Performance depends on Δ
  - if Δ too small will not encompass entire locality
  - if Δ too large will encompass several localities
  - if $\Delta = \infty \Rightarrow$ will encompass entire program

Idle page time (T)

- Pages with idle time less than T are in the working set.

Working-Set model prevents thrashing while keeping the degree of multiprogramming as high as possible.

Page-Fault Frequency Scheme

- Establish "acceptable" page-fault rate
  - If actual rate too low, process loses frame
  - If actual rate too high, process gains frame